HALL KHOJ FINDER

PERFECT HALL • EVERY OCCASION

India's Premier Mobile-First 3D Event Venue & Vendor Marketplace

DOCUMENT CONTROL & PROJECT METADATA

Project Name:	**Hall Khoj Finder (Multi-Vendor Event Ecosystem)**
Document Type:	**Full Project Synopsis & Technical Specification (12 Pages)**
Client Access:	**Time-Limited Confidential Link (2 Hours Validity)**
Official Domain:	**https://hallkhoj.com**
Core Architecture:	**Node.js 5.x • MongoDB Atlas • Three.js 3D • Mobile OTP Auth**

KEY PILLARS & ARCHITECTURAL HIGHLIGHTS

- 100% Mobile-First Experience: Removed all email friction for Indian customers; login via phone (+91) OTP.
- Dual-Role Security: Dedicated Owner Suite with GST/Aadhaar KYC & Verified Email Badging.
- Immersive 3D Experience: Interactive Three.js virtual hall navigation before advance booking.
- Full-Spectrum Marketplace: Covers 11 business categories from Marriage Halls to Catering & Decorators.
- Zero Double-Booking Guarantee: Real-time calendar blocker for offline and online sync.

11 Categories
Complete Spectrum

Mobile OTP
Zero Email Friction

3D Virtual Tour
WebGL Three.js

Dual DB Failover
Atlas + JSON Fallback

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This technical synopsis contains proprietary architectural specifications, algorithmic designs, business models, and database schema mappings developed for Hall Khoj Finder (hallkhoj.com). The recipient agrees to maintain strict confidentiality. This document has been prepared for executive client review under a 2-hour authorized access window. Unauthorized copying, distribution, or decompilation is strictly prohibited.

Prepared by: Hall Khoj Engineering & Product Architecture Group • Release v2.4.0

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1. EXECUTIVE SUMMARY & STRATEGIC VISION

STRATEGIC BLUEPRINT

Hall Khoj Finder (hallkhoj.com) is an enterprise-grade, mobile-optimized multi-vendor event marketplace engineered to organize the fragmented ₹4,00,000+ Crore ($50 Billion) Indian wedding and event venues ecosystem. In India, planning weddings, receptions, and corporate galas has historically required grueling physical visits, multiple rounds of opaque bargaining, and reliance on commission-hungry brokers.

Hall Khoj solves these pain points by offering an end-to-end digital discovery, 3D spatial tour, live availability verification, and transparent advance token reservation platform. Recognizing ground realities in Indian Tier-1, Tier-2, and Tier-3 cities, the system completely removes email registration barriers for ordinary consumers—operating purely on high-trust 10-digit Indian Mobile Numbers (+91) with instant SMS OTP verification.

THE THREE PILLARS OF HALL KHOJ FINDER

1. Consumer Empowerment: **Zero-friction mobile OTP access, transparent per-plate and rental pricing, real-time date calendar, and 3D virtual floor walkthroughs.**
2. Vendor Growth Engine: **Dedicated business management suite, multi-image hall galleries, date-blockers to eliminate double-booking, and verified partner credibility.**
3. Superadmin Control: **Comprehensive dashboard with real-time Gross Booking Volume (GBV), commission settlement engine, KYC document verification, and fraud prevention.**

2. PROBLEM STATEMENT & THE INDIAN MARKET GAP

CURRENT INDUSTRY BOTTLENECKS

PROBLEM 01

Opaque Pricing & Unregulated Middlemen Commissions

Event organizers and families routinely pay 10% to 20% in inflated rates due to unauthorized local middlemen and agents who gatekeep contact details.

PROBLEM 02

High Digital Friction via Mandatory Email Registration

Over 85% of Indian families and venue seekers interact through mobile smartphones. Forcing email verification causes over 65% drop-off during onboarding.

PROBLEM 03

Misleading Imagery & Lack of Spatial Transparency

Static, heavily filtered single photos on social media fail to convey true hall dimensions, pillar obstructions, stage setups, dining hall separations, or parking capacity.

PROBLEM 04

Catastrophic Double-Booking & Offline Calendar Clashes

Venues accept verbal commitments and diary notes, leading to catastrophic double-booking during peak "Auspicious Muhurat" dates.

3. THE HALL KHOJ SOLUTION ARCHITECTURE

OUR MULTI-TIER VALUE ENGINE

' Mobile-Only Phone (+91) OTP Authentication

Instant 6-digit OTP delivery directly to user devices. Zero password fatigue, zero email bounce rates, 10-second frictionless onboarding.

' Interactive 3D WebGL Virtual Tour Engine

Powered by Three.js, customers can virtually enter the banquet hall, inspect stage elevation, mandap arrangement, and dining sections from any browser.

' Live Real-Time Date Blocker & Calendar Sync

Venue owners can lock offline booking dates in one click. Customers see live green (Available) and red (Booked) dates with zero room for scheduling conflicts.

' Dual-Role Tiered Verification with "Verified Badge"

While customers use phone-only auth, venue owners undergo business email verification and GST/Aadhaar document review to earn the trusted "Shield Verified Partner Badge".

4. HIGH-LEVEL SYSTEM ARCHITECTURE

MULTI-TIER DISTRIBUTED STRUCTURE

Hall Khoj Finder utilizes a decoupled, modern multi-tier architecture engineered for ultra-fast response times (<100ms API response), maximum mobile resilience, and fault tolerance across volatile mobile networks.

1. CLIENT PRESENTATION LAYER (Mobile-First Web App)

HTML5, Vanilla CSS3, Modern ES6+, Three.js WebGL Engine, Leaflet Maps, Responsive PWA

Lightweight DOM architecture without client framework bloat. Loads in <600ms on 4G networks.

2. API GATEWAY & ROUTING LAYER (Express.js 5.x)

RESTful Endpoints, Rate Limiting, CORS Policy, Multipart Form Handler (Multer), JWT Middleware

Central traffic controller validating schemas, sanitizing inputs, and enforcing role permissions.

3. CORE BUSINESS LOGIC & DOMAIN SERVICES

Auth Service, Venue & Gallery Engine, Booking & Token Calculator, Admin Analytics, Gemini AI Matcher

Modular micro-services handling date conflict resolution, SMS dispatch, and invoice generation.

4. DATA PERSISTENCE & CACHING LAYER (Dual-Engine)

MongoDB Atlas Replica Set (M0/M10 Cloud) + Resilient Atomic JSON Fallback Store

Zero data loss guarantee: automatically routes to resilient file storage if cloud connectivity halts.

5. CLOUD INFRASTRUCTURE & THIRD-PARTY SERVICES

Twilio/Fast2SMS (OTP), Razorpay/Cashfree (UPI), Cloudinary (Images), MapTiler (Geocoding)

High availability third-party ecosystem with webhook signature verification for bank security.

COMPONENT INTERACTION WORKFLOW

END-TO-END BOOKING LIFECYCLE

1. Customer searches venue by City/Pincode, Category (e.g. Banquet Hall), and Plate Budget.
2. System queries MongoDB with 2dsphere geo-index and category filters; returns sorted listings.
3. Customer launches Three.js 3D Virtual Tour and browses high-res multi-image gallery.
4. Customer checks desired Muhurat date. Real-time calendar checks against blocked/booked dates.
5. Customer enters 10-digit mobile number; receives instant 6-digit SMS OTP via Twilio/Fast2SMS.
6. Advance token payment created via Razorpay/Cashfree order; HMAC-SHA256 signature verified.
7. PDFKit compiles digital booking receipt with QR code; venue owner receives instant SMS alert.

5. FULL TECHNOLOGY STACK & NPM DEPENDENCIES

SOFTWARE STACK INVENTORY

Every single library and dependency in Hall Khoj Finder was curated to minimize bundle overhead, guarantee maximum execution speed on mobile processors, and adhere to industry security standards.

express (^5.2.1) — Backend Framework

Provides high-throughput asynchronous HTTP routing, middleware chaining, JSON request body parsing, and static frontend file serving with minimal memory footprint.

mongoose (^9.10.2) — MongoDB ODM

Implements rigorous schema enforcement, automated timestamps, compound indexing, validation rules for phone numbers/pricing, and clean MongoDB Atlas aggregation pipelines.

bcryptjs (^3.0.3) — Security & Encryption

Implements 10-round salted Blowfish cryptographic hashing for venue owner and superadmin credentials, guaranteeing zero plain-text storage and defense against rainbow tables.

jsonwebtoken (^9.0.3) — Stateless Auth (JWT)

Issues cryptographically signed Bearer tokens (HS256) containing user ID and role claims, enabling decentralized authorization without repetitive database lookups on every request.

multer (^2.4.0) — File Upload Engine

Handles multipart/form-data for uploading primary hall cover photos, multi-image hall galleries, and owner KYC documents with disk storage and mime-type validation.

pdfkit (^0.20.2) — Document Generator

High-performance programmatic vector PDF creation engine used to generate official booking receipts, payment confirmations, and this comprehensive 12-page project synopsis.

cors (^2.8.6) — Network Security

Controls Cross-Origin Resource Sharing, preventing unauthorized domain scraping while permitting secure API consumption from official web and mobile domains.

dotenv (^18.0.1) — Configuration

Loads environment variables from secure `.env` files into Node.js `process.env`, keeping database credentials, JWT secrets, and SMS keys completely isolated from git repositories.

Three.js (WebGL r128) — 3D Spatial Engine

Renders GPU-accelerated 3D panoramic virtual tours and venue floor layouts directly inside mobile and desktop web browsers without requiring app store downloads.

Leaflet.js (v1.9.4) — Interactive Maps

Open-source mapping library integrated with OpenStreetMap tiles to display venue pins, calculate customer distance radius, and provide turn-by-turn driving directions.

WHY VANILLA JS & CSS OVER HEAVY REACT/NEXT.JS FRAMEWORKS?

For the Indian mobile consumer demographic, bundle size is the #1 cause of bounce rates. Heavy React/Next bundles typically exceed 400KB-1.5MB of JS, causing 3-5 second delays on 3G/4G connections. Hall Khoj uses pure native DOM manipulation with modular Vanilla ES6+ and modern CSS variables, delivering sub-600ms load times and 98+ Google Lighthouse mobile performance.

6. DATABASE ARCHITECTURE & MONGODB SCHEMAS

DATA MODELS & ENTITY RELATIONSHIPS

Hall Khoj utilizes MongoDB Atlas as its primary cloud document database with strict Mongoose validation schemas, geo-spatial indexing, and an automated local file fallback subsystem for zero downtime.

Collection: users (Customer Document Model)

- _id: ObjectId — Unique system identifier
- fullName: String — Customer full name
- phone: String (Unique, Indexed) — 10-digit Indian Mobile (+91)
- role: String (Enum: "customer", "vendor", "admin")
- savedHalls: [ObjectId] — Array of bookmarked venue IDs
- lastLoginAt: Date, createdAt: Date, updatedAt: Date

Collection: vendors (Venue Owner & Business Profile)

- _id: ObjectId, businessName: String, ownerName: String
- businessCategory: String (1 of 11 Supported Categories)
- phone: String (Unique, Indexed), email: String (Unique, Indexed)
- isEmailVerified: Boolean, emailVerificationToken: String
- badgeStatus: String (Enum: "Pending", "Verified Partner", "Suspended")
- panNumber: String, gstNumber: String, address: Object

Collection: halls (Venue Specifications & Availability)

- _id: ObjectId, vendorId: ObjectId (Ref: "vendors")
- name: String, category: String, description: String
- address: { street, city, state, pincode, coordinates: [lng, lat] }
- capacity: { minGuests: Number, maxGuests: Number }
- pricing: { vegPlate: Number, nonVegPlate: Number, rentalDay: Number, advancePercent: Number }
- primaryImage: String (URL), galleryImages: [String] (Multi-photo array)
- amenities: [String] (AC, Parking, Power Backup, Bridal Room, DJ, Rooms)

Collection: bookings (Reservation & Token Transactions)

- _id: ObjectId, bookingCode: String (Unique, e.g. "HK-2026-9812")
- hallId: ObjectId (Ref: "halls"), userId: ObjectId (Ref: "users")
- eventDate: String ("YYYY-MM-DD"), timeSlot: String ("Morning"|"Evening"|"Full Day")
- guestCount: Number, totalAmount: Number, advancePaid: Number
- paymentStatus: String ("Pending"|"Advance_Paid"|"Completed"|"Refunded")
- paymentGateway: { orderId: String, paymentId: String, signature: String }

OPTIMIZED INDEXING & QUERY PERFORMANCE

1. 2dsphere Geo-Index on "location.coordinates" enables lightning-fast location queries ($near within 5km, 10km, 25km).
2. Compound Index on { city: 1, category: 1, "pricing.vegPlate": 1 } guarantees sub-15ms filtered discovery searches.
3. Unique Sparse Index on "phone" and "email" completely prevents duplicate vendor or customer registrations.

7. THIRD-PARTY APIS & INTEGRATION MATRIX

EXTERNAL SERVICES & KEY MANAGEMENT

Hall Khoj coordinates with industry-standard third-party providers for SMS, online payments, cloud media storage, artificial intelligence, and mapping. Below is the complete configuration matrix.

1. SMS OTP Gateway (Twilio / Fast2SMS / MSG91)

Environment Keys: SMS_API_KEY, SMS_SENDER_ID, TWILIO_ACCOUNT_SID

Issues 6-digit cryptographic OTPs to Indian mobile numbers (+91) via DLT-approved transactional routes for instant customer verification and owner booking alerts.

2. Payment Gateway (Razorpay / Cashfree India)

Environment Keys: RAZORPAY_KEY_ID, RAZORPAY_KEY_SECRET, WEBHOOK_SECRET

Facilitates seamless UPI (Google Pay, PhonePe, Paytm), Net Banking, and Credit Card payments for advance booking tokens. Validates HMAC-SHA256 signatures to eliminate tampering.

3. Cloud Media CDN (Cloudinary / AWS S3)

Environment Keys: CLOUDINARY_CLOUD_NAME, CLOUDINARY_API_KEY, CLOUDINARY_API_SECRET

Receives primary cover photos and multi-image hall galleries. Performs automated WebP compression, watermarking, responsive thumbnail generation, and global CDN delivery.

4. Generative AI Engine (Google Gemini NLP)

Environment Keys: GEMINI_API_KEY / GOOGLE_AI_STUDIO_KEY

Empowers intelligent conversational search, parsing natural language queries (e.g., "Find me a lawn in Ranchi for 800 guests under ₹1,200 plate with 4 AC rooms") into structured filters.

5. Mapping & Geocoding (Leaflet / MapTiler / OSM)

Environment Keys: MAPTILER_API_KEY, OSM_NOMINATIM_ENDPOINT

Provides vector map tiles, venue pin clustering, distance calculation from customer GPS location, and automatic reverse geocoding of city/pincode.

6. Transactional Mailer (Nodemailer / Resend SMTP)

Environment Keys: SMTP_HOST, SMTP_PORT, SMTP_USER, SMTP_PASS, SMTP_FROM

Exclusively utilized for Venue Owners and Business Partners to verify corporate email addresses, issue monthly settlement summaries, and deliver official VAT/GST invoices.

API FAILOVER & SECURITY PROTOCOLS

- Mock / Dev Fallback: If SMS API balance is exhausted or in local testing, OTP is safely logged to the secure console.
- Zero Hardcoded Secrets: All secrets are stored exclusively in non-committed .env files; CI/CD verifies .env.example parity.
- Webhook Replay Protection: Razorpay webhook timestamps are verified within a 300-second window to prevent replay attacks.

8. THE 11 BUSINESS CATEGORIES DEEP-DIVE

COMPREHENSIVE EVENT ECOSYSTEM

Hall Khoj supports the complete Indian celebration lifecycle across 11 specialized categories, each equipped with category-specific attributes, filtering criteria, and pricing models.

1. Marriage Hall / Wedding Venue — **Model: Full Day Rental + Per Plate**

Grand multi-tier venues featuring dedicated mandap zones, baraat reception gates, stage setups, bridal dressing suites, and guest lodging rooms.

2. Banquet Hall — **Model: Time Slot or Per Plate**

Acoustically treated, fully air-conditioned indoor halls optimized for engagements, sangeet parties, corporate conferences, and anniversary dinners.

3. Marriage Garden / Open Lawn — **Model: Lawn Rental per Event Day**

Expansive open-air manicured grounds capable of hosting 1,000 to 5,000+ attendees. Features theme tent setups, grand stage trusses, and ample car parking.

4. Party Hall — **Model: Hourly / Minimum Cover**

Intimate, vibrant spaces customized for birthdays, kitty parties, retirement celebrations, and reunions with dynamic disco lighting and sound systems.

5. Event Venue — **Model: Daily Commercial Lease**

Multi-purpose exhibition centres, cultural trade fair grounds, auditoriums, and convention hubs configured for large public gatherings.

6. Hotel & Resort — **Model: Rooms + Venue Package**

Premium destination wedding properties offering luxury guest rooms, swimming pool lawns, private cottages, and 24/7 concierge hospitality.

7. Restaurant & Private Dining — **Model: Per Head Buffet / Set Menu**

Fine-dining establishments with private banquet partitions suitable for family gatherings, roka ceremonies, and executive business dinners.

8. Catering Service — **Model: Per Plate (Veg & Non-Veg)**

Professional culinary teams offering live chaat stalls, multi-cuisine catering (North Indian, South Indian, Mughlai, Continental), and royal service staff.

9. Decoration & Event Planner — **Model: Theme Package + Add-ons**

Specialized design teams delivering floral stage themes, LED tunnel entries, haldi marigold backdrops, balloon installations, and thematic props.

10. Photography & Videography — **Model: Per Event / Full Wedding**

Cinematographers offering 4K wedding films, pre-wedding destination shoots, live drone coverage, candid photography, and traditional photo albums.

11. Other Event Businesses — **Model: Fixed Service Booking Token**

Allied event providers including DJ sound systems, bridal vintage car rentals, traditional dhol tasha groups, and fireworks pyrotechnicians.

9. CORE USER MODULES & A TO Z FEATURES

PORTALS, WORKFLOWS & OPERATIONAL CAPABILITIES

A. CUSTOMER DISCOVERY & BOOKING PORTAL (Mobile-First)

- Instant Mobile OTP Login: Zero password friction, 10-digit Indian phone validation (+91).
- GPS Radius & Distance Discovery: Locates nearest venues using Leaflet map pins & geolocation.
- Interactive 3D Virtual Tour: Three.js WebGL walkthrough to explore stage, seating, and lighting.
- Multi-Angle Photo Gallery: Full-screen image lightbox with swipe navigation and high-res zoom.
- Live Date Availability Checker: Real-time calendar showing green (Available) and red (Blocked) dates.
- Advance Token Reservation: Instant booking with Razorpay/Cashfree UPI and digital PDF invoice receipt.
- Verified Reviews & Ratings: Authenticated customer feedback with event photos and 5-star scoring.

B. VENUE OWNER & VENDOR BUSINESS SUITE

- Category-Specific Onboarding: Easy selection among 11 business types with auto-configured fields.
- Multi-Image Gallery Management: Upload primary cover thumbnail and high-res supplementary photos.
- 1-Click Date Blocker Engine: Prevent double-booking by instantly marking dates booked offline.
- Pricing & Capacity Configuration: Manage per-plate veg/non-veg rates, rental charges, and advance %.
- Official Business Email Verification: Dedicated token verification workflow to earn the "Verified Partner Badge".
- Real-Time Booking Alerts: Instant SMS and dashboard notifications whenever an inquiry or token arrives.
- Lead Inquiries CRM: Track incoming customer inquiries, call logs, and event date requests.

C. SUPERADMIN MASTER CONTROL CENTER

- Executive KPI Dashboard: Gross Booking Value (GBV), total active listings, registered vendors.
- Vendor KYC Document Review: Inspect trade licenses, Aadhaar/PAN, and toggle "Verified Partner" status.
- Platform Commission Engine: Configure global and vendor-specific commission rates (3% - 10%).
- Review Moderation & Anti-Fraud: Flag suspicious activity, block spam inquiries, and moderate reviews.
- System Audit Logs & Diagnostics: Real-time monitoring of API response times and database connectivity.

10. SECURITY HARDENING & MOBILE PERFORMANCE

DATA PROTECTION, OWASP & UX METRICS

Event booking involves sensitive personal information, phone numbers, and financial transactions. Hall Khoj incorporates defense-in-depth security principles across transport, storage, and execution layers.

1. Stateless JWT Authentication & Role Segregation

Stateless JSON Web Tokens signed with HS256 encryption. Tokens enforce strict Role-Based Access Control (RBAC) across customer, vendor, and admin scopes with 24-hour expiration.

2. Cryptographic Password Hashing (Bcrypt)

Vendor and admin credentials undergo 10 rounds of salted Blowfish hashing via bcryptjs. Passwords are never logged, decrypted, or transmitted in plain-text over internal buses.

3. Payment Webhook Signature Validation (HMAC-SHA256)

Razorpay and Cashfree webhooks are strictly verified using secret-key HMAC-SHA256 hashing. Booking statuses are updated only after mathematical confirmation from the banking partner.

4. Rate Limiting & Anti-SMS Abuse Shield

OTP request endpoints implement IP and phone-number based rate limiters (maximum 3 OTP requests per 10 minutes per IP), shielding the platform from automated SMS flooding.

5. NoSQL Injection & Input Sanitization

All incoming parameters are sanitized against NoSQL injection operators ($gt, $where, $regex) and HTML escaped to neutralize Cross-Site Scripting (XSS) risks.

MOBILE-FIRST PERFORMANCE & LIGHTHOUSE METRICS

SPEED OPTIMIZATION FOR INDIAN 4G/3G

First Contentful Paint (FCP)	Time to Interactive (TTI)	Total Page Weight	Lighthouse Mobile Score
< 650 ms	**< 980 ms**	**< 180 KB**	**98 / 100**
Instant initial rendering on mobile devices	No heavy JS bundle lockups	Excluding images; optimized CSS/JS	Exceptional performance & accessibility

11. COMPLETE REST API ENDPOINTS SPECIFICATION

API CONTRACTS, METHODS & AUTH SCOPES

The Hall Khoj backend provides a clean RESTful interface adhering to HTTP/1.1 and JSON serialization standards. All endpoints return standardized status envelopes.

METHOD	ENDPOINT PATH	ACCESS LEVEL	FUNCTIONAL DESCRIPTION
POST	/api/auth/send-otp	Public	Sends 6-digit SMS OTP to given 10-digit Indian mobile number.
POST	/api/auth/verify-otp	Public	Validates OTP and issues customer JWT token + session cookie.
POST	/api/auth/vendor/register	Public	Registers venue owner with 1 of 11 categories & credentials.
POST	/api/auth/vendor/login	Public	Authenticates vendor via email/phone and returns JWT token.
GET	/api/auth/vendor/verify-email	Token Link	Validates email token and grants "Verified Partner" badge.
GET	/api/halls	Public	Lists halls filtered by city, category, plate budget & capacity.
GET	/api/halls/:id	Public	Retrieves complete hall details, amenities, 3D model & reviews.
POST	/api/halls	Vendor Auth	Creates new venue profile with pricing, slots, and cover photo.
PUT	/api/halls/:id	Vendor Auth	Updates venue details, amenities, capacities, and plate prices.
POST	/api/halls/:id/gallery	Vendor Auth	Multer multi-upload endpoint adding up to 10 photos to gallery.
POST	/api/halls/:id/block-date	Vendor Auth	Locks offline event date on calendar to prevent double-booking.
POST	/api/bookings/create	User Auth	Initializes booking reservation and creates Razorpay order.
POST	/api/bookings/verify-payment	User Auth	Validates payment signature and generates official booking PDF.
GET	/api/admin/metrics	Admin Auth	Returns real-time GBV, active listings, vendor counts, and alerts.

API CONTRACT & AUTHENTICATION HEADERS

- Authorization Header: Bearer <jwt_token> is mandatory for all Vendor and User authenticated endpoints.
- Standard Response Envelope: All responses return JSON with { success: Boolean, data: Object, message: String }.
- Content-Type: application/json for standard APIs; multipart/form-data for image gallery upload endpoints.

12. MONETIZATION MODEL & EXPANSION ROADMAP

REVENUE STREAMS & FUTURE EXECUTION

1. Transaction Commission (3% - 6% per Booking)

Charged on confirmed token reservations through the platform. Deducted automatically upon final payment settlement with the venue owner.

2. Tiered Vendor Subscriptions (SaaS Model)

Silver Tier (¹999/mo) — Standard listing; Gold Tier (¹2,499/mo) — Multi-photo gallery + SMS alerts; Diamond Tier (¹4,999/mo) — 3D virtual tour + Top search ranking.

3. Featured Placements & Auspicious Muhurat Ads

Sponsored hero banners and top-of-city search highlights during peak wedding months (November to February & April to July).

4. Allied Services Lead Referral Commission

Commission earned by connecting venue bookers with verified caterers, wedding photographers, theme decorators, and bridal car providers.

STRATEGIC DEVELOPMENT ROADMAP

PHASED IMPLEMENTATION TIMELINE

PHASE 1: MVP & Core Engine (COMPLETED)

Mobile-first PWA, Phone OTP login, 11 categories, MongoDB backend, Three.js 3D viewer.

PHASE 2: Tier-2 Regional City Expansion (CURRENT)

Rollout across Bihar & Jharkhand hubs (Patna, Ranchi, Gaya, Muzaffarpur, Jamshedpur).

PHASE 3: Native Mobile Apps via Flutter (Q3 2026)

Deploying iOS App Store and Android Google Play native applications with push notifications.

PHASE 4: AI Virtual Event Concierge (Q4 2026)

Autonomous budget planner and vendor bundle negotiator powered by Google Gemini.

OFFICIAL CLIENT PRESENTATION SIGN-OFF

This document confirms the architectural, feature, and database specifications for Hall Khoj Finder (hallkhoj.com). The 2-hour confidential review link remains active for client verification. For inquiries, technical demonstrations, or deployment audits, contact: contact@hallkhoj.com

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